



CMS CAPILLARIES PRODUCTION FOR LHC HIGH LUMINOSITY UPGRADE

Konrad Szczyrbak | Arianna Eugeni

Contribution: Jérôme Daguin, Derek Langedijk & Lisa Gall

Forum on Tracking Detector Mechanics 2026



CAPILLARIES IN CMS (Compact Muon Solenoid)

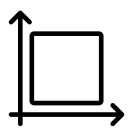
With the **High Luminosity** LHC delivering up to 7.5 times more luminosity than the current LHC, CMS will upgrade its detectors to cope with harsher radiation, higher data rates, and increased pileup. To operate safely under these conditions, the upgraded silicon detectors require a highly efficient cooling system: since the increased radiation levels induce higher leakage currents in the sensors, the modules must be kept at **-35°C** to prevent thermal runaway. **Bi-phase CO₂** cooling is employed to meet these strict thermal requirements, operating on a **2PACL** (2-Phase Accumulator Controlled Loop) cycle.

Within this 2PACL system, capillaries ensure :

- The **correct flow distribution** in each of the parallel branches of sub-detectors: each capillary guarantees about 80% of the total pressure drop of a detector loop, assuring that the neighboring loops can still function in case of circuit pressure unbalance.
- A well-defined pressure drop at the detector inlet, which is crucial for efficient cooling. The aim is to **trigger boiling** inside the evaporator of the detector, resulting in an improved heat transfer.

Parameter	Typical value
Outer diameter (OD)	1.2 mm
Inner diameter (ID)	0.5 - 0.8 mm
Typical length	400- 2000 mm
Material	AISI 316L
Operating pressure	Approx. 35 bar
Temperature range	20°C to -40°C
Number of capillaries	2094

ENGINEERING CHALLENGES



SMALL DIMENSIONS

Precision handling of 1.2 mm capillaries



LEAK TEST

Ensuring vacuum and pressure tightness



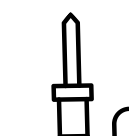
CAPILLARIES PURITY

Preventing internal contamination



TEMPERATURE

Maintaining -35°C temp. during testing



BRAZING

Ensuring process repeatability while reducing failures

PRODUCTION PROCESS

RAW MATERIAL

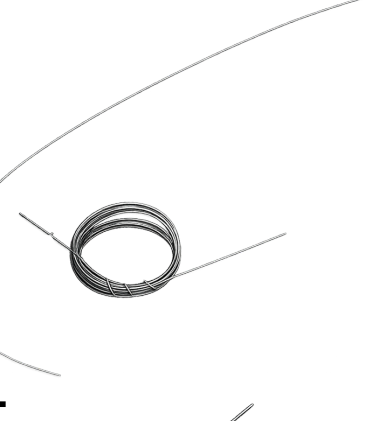
Stainless steel pipes are supplied in large spools with an outer diameter of 1.2 mm and inner diameters ranging from 0.5 to 0.8 mm. Each spool is subjected to a metrological inspection to verify ID and OD tolerances.



PROTOTYPING

Capillaries are manually cut from the spool, and both ends are polished to fully open the inner channel.

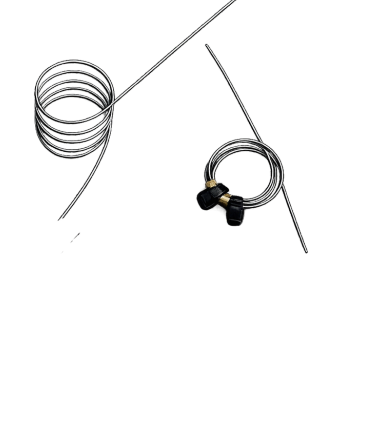
They are then formed into the final detector geometry to validate the required shape and length.



EDM CUT

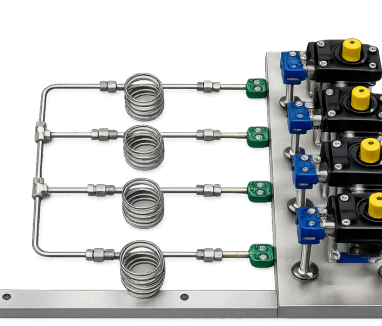
After defining the optimal geometry and length, the required number of capillaries is manually prepared with an additional +100 mm margin.

The capillaries are grouped into batches and transferred for precision EDM cutting to achieve smooth and repeatable end surfaces.



TESTING

EDM-cut capillaries are installed in the COS (Capillary Open System) testing system and operated under controlled mass flow and temperature conditions. During the test, pressure drop (DP) stability is monitored. A stable operating period of 5 minutes is required for data validation. Each capillary has its own metal tag with a unique identification number.



BRAZING

Validated capillaries are joined to the 1/8" VCR connectors via furnace or induction vacuum brazing (BFM: Nicrobraz 4778; maximum temperature cycle: 1060°C).

Since the connectors are suspended during brazing, the capillary ends are gently pre-bent to avoid any relative movement, ensuring a 2mm protrusion beyond the gland surface.



JOINTS QUALITY CONTROL

Each assembled capillary undergoes pressure and leak testing to verify mechanical integrity and tightness. When required, X-ray inspection is additionally performed to validate the internal quality of the brazed joints and detect possible defects. Only components fulfilling acceptance criteria are approved for detector integration.



DATA MANAGEMENT

All production parameters, testing results, brazing records and quality control measurements are stored in the CERN EDMS database.

This ensures full traceability and long-term production monitoring for each capillary batch.



COILING IMPACT ON DP

CAP NAME	GEOMETRY	ID (mm)	L (mm)	NUMBER OF LOOPS	FLOW (gr/s)	DP (bar)	DP/flow	DP difference	DP increase per loop (bar)
HGCAL A	STRAIGHT	0,80	1250	14 loops	3,50	9,64	2,75	1,69 ↑	0,12
	COILED				3,51	11,33	3,23		
TBPX MT Z-L4-10	STRAIGHT	0,80	1000	10 loops	3,53	7,46	2,10	1,45 ↑	0,15
	COILED				3,54	8,91	2,52		
TEDD 1	STRAIGHT	0,60	890	8 loops	1,86	8,75	4,71	1,35 ↑	0,16
	COILED				1,86	10,10	5,42		
TEDD 2	STRAIGHT	0,60	1275	14 loops	1,59	8,84	5,56	1,72 ↑	0,12
	COILED				1,59	10,56	6,63		
TBPX MT Z-L2-8	STRAIGHT	0,50	860	6 loops	1,29	9,38	7,26	1,37 ↑	0,20
	COILED				1,29	10,74	8,32		
TBPX MT Z-L1-4	STRAIGHT	0,50	840	6 loops	1,28	9,31	7,26	1,21 ↑	0,23
	COILED				1,28	10,52	8,20		

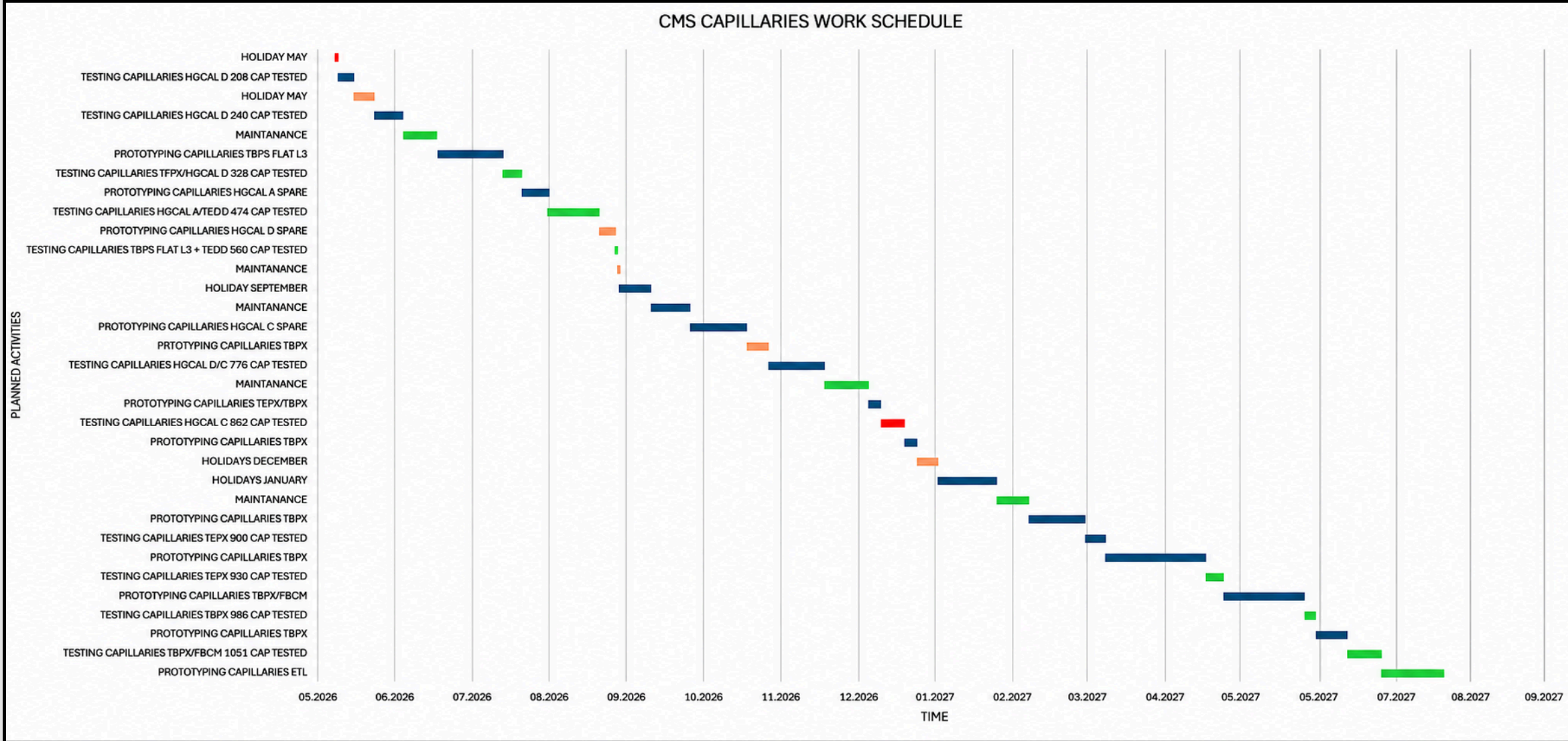
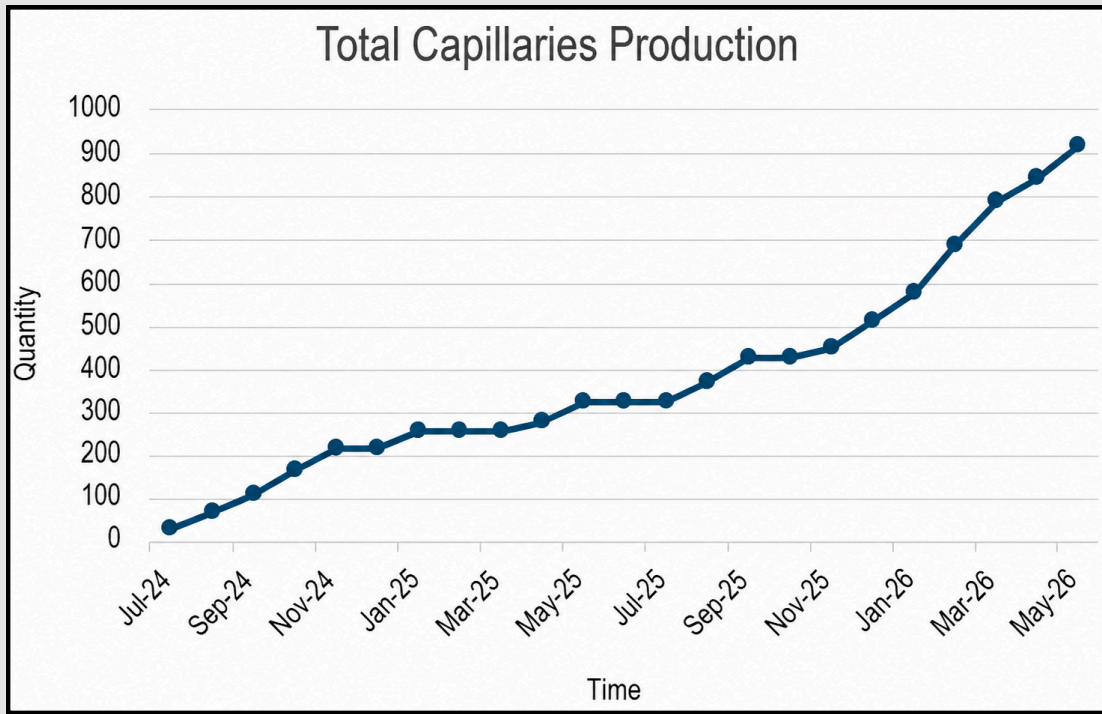
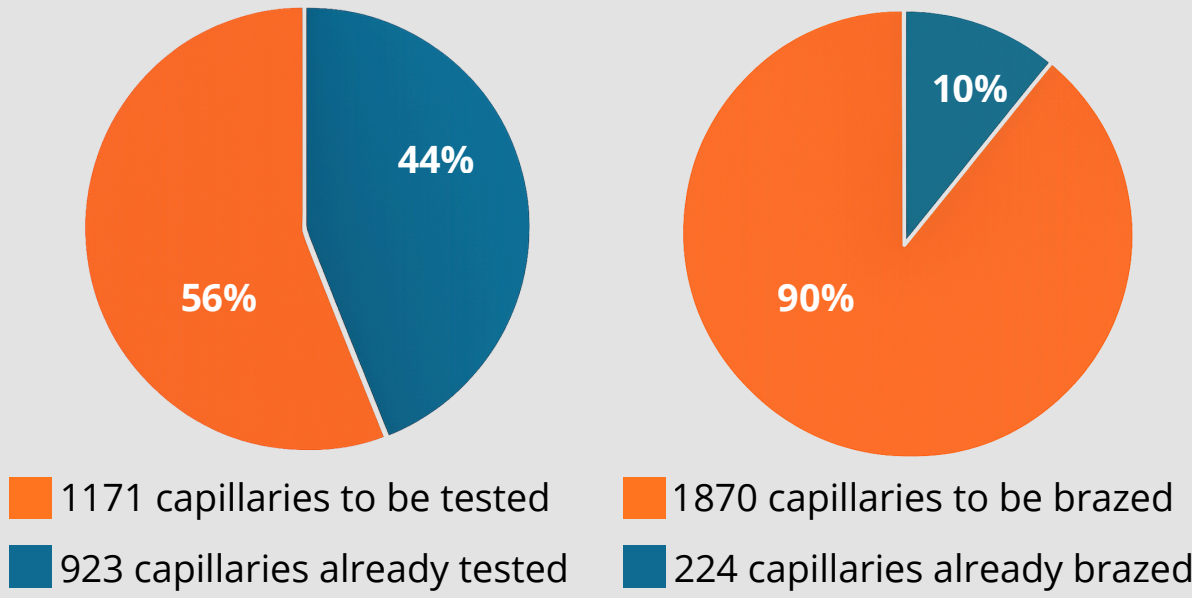
RESULTS & OUTCOME

DP sensor error: +/-0,05%
Flow meter sensor error: +/-0,20%

CMS SUB-DETECTRS CAPILLARIES OVERVIEW						
NAME	INPUT DATA		TEST DATA		ADDITIONAL VALUES	
	FLOW (g/s)	DP (bar)	FLOW (g/s)	DP (bar)	DP st.dev (bar)	FLOW st.dev (g/s)
*BTL	2,65	9,45	2,60	9,46	0,02	0,02
*HGCAL A	3,50	9,50	3,46	9,50	0,02	0,04
HGCAL B	4,00	9,50	4,10	9,51	0,03	0,03
*HGCAL C	2,30	9,50	2,28	9,51	0,03	0,01
TB2S	1,26	8,80	1,26	8,79	0,12	0,05
TBPS TILTED L1	1,61	9,10	1,62	9,11	0,02	0,02
TBPS TILTED L2	2,16	8,10	2,17	8,09	0,04	0,02
TBPS TILTED L3	2,83	8,40	2,85	8,40	0,02	0,04
*TEDD 1	1,88	8,67	1,88	8,67	0,02	0,03
*TEDD 2	1,59	8,83	1,54	8,83	0,01	0,01

* ongoing process of production

TBPX, TFPX, TBPS FLAT L1, TBPS FLAT L2 - prototyping phase



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Isola d'Elba, Italy



CMS capillaries EDMS folder
<https://edms.cern.ch/project/CERN-0000270811>